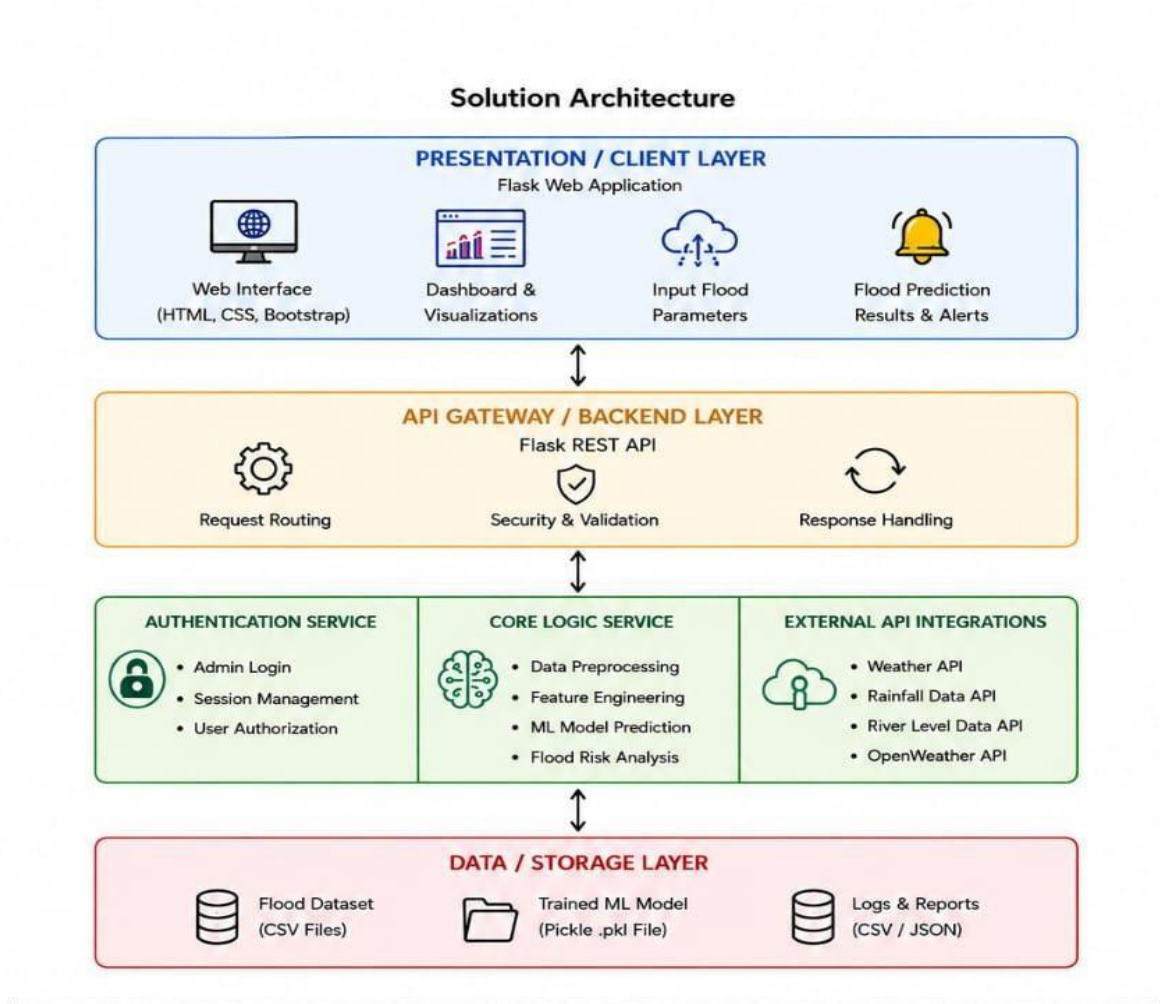


Solution Architecture

Date	06 July 2026
Project Title	Rising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram :



Component Description Table :

Component Name	Description / Role in Architecture	Technology used
Presentation Layer	User interface to enter flood parameters and view prediction results	Flask, HTML, CSS, Bootstrap
API Gateway	Receives user requests and sends them to the ML model	Flask REST API, Python
Authentication	Handles admin login and user sessions	Flask Session
Core logic service	Data preprocessing, feature extraction, machine learning prediction	Python, Pandas, NumPy, Scikit-learn
External API	Fetches weather and rainfall information	Open Weather API (optional)
Database / Storage	Stores dataset and trained model	CSV Dataset, Pickle (.pkl)